

HYDRO TESTING UNIT.

Programmed By :- Somnath Vishwakarma



PLC Input Output Diagram

INPUTS

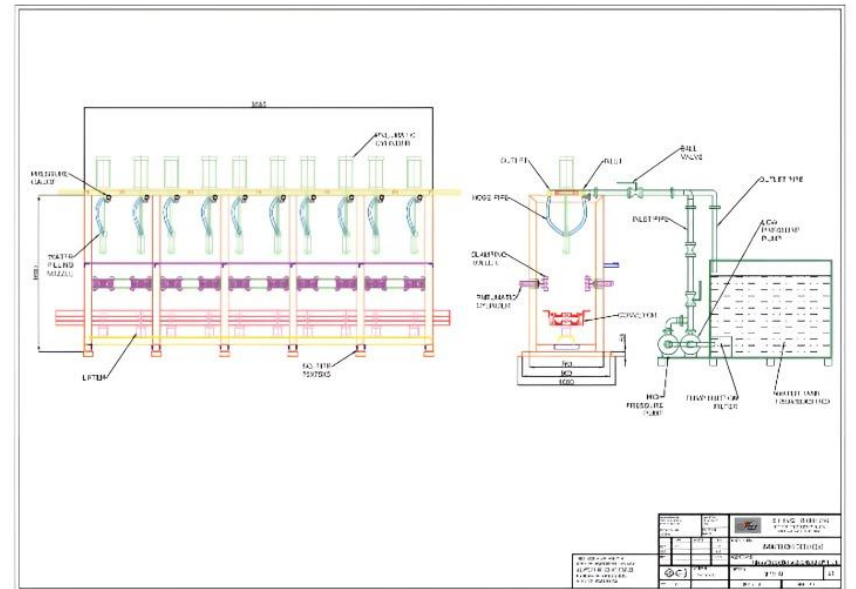
X0	BYPASS_HEAD_05
X1	BYPASS_HEAD_09_10
X2	STOP
X3	BYPASS_HEAD_07_08
X4	BYPASS_HEAD_06
X5	START
X6	EMG_SWITCH
X20	BYPASS_HEAD_03
X21	BYPASS_HEAD_02
X22	BYPASS_HEAD_01
X23	BYPASS_HEAD_04
X24	Tilting Limit Switch
X25	Home Position Limit Switch
X26	Pressure Switch

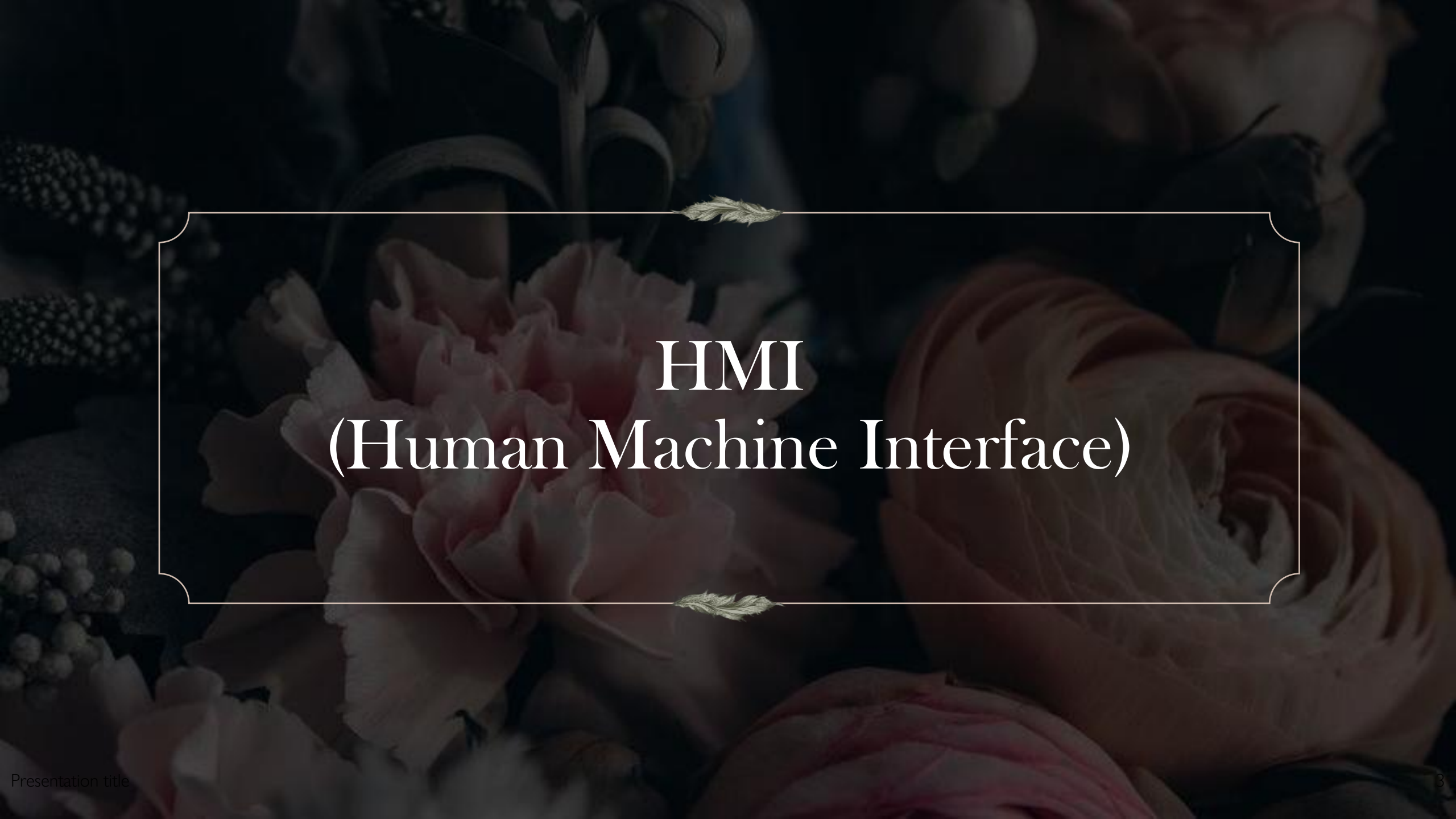
OUTPUTS

Y0	Head_06
Y1	Head_10
Y2	Head_03
Y3	Head_07
Y4	Head_02
Y5	Head_04
Y20	Head_08
Y21	Head_09
Y22	Head_05
Y23	Head_01
Y24	Tilting
Y25	Gun_down
Y26	Red Light
Y27	Main Filling ROV
Y30	High Pressure Pump
Y31	Air_ROV
Y32	Drain_ROV

OUTPUTS

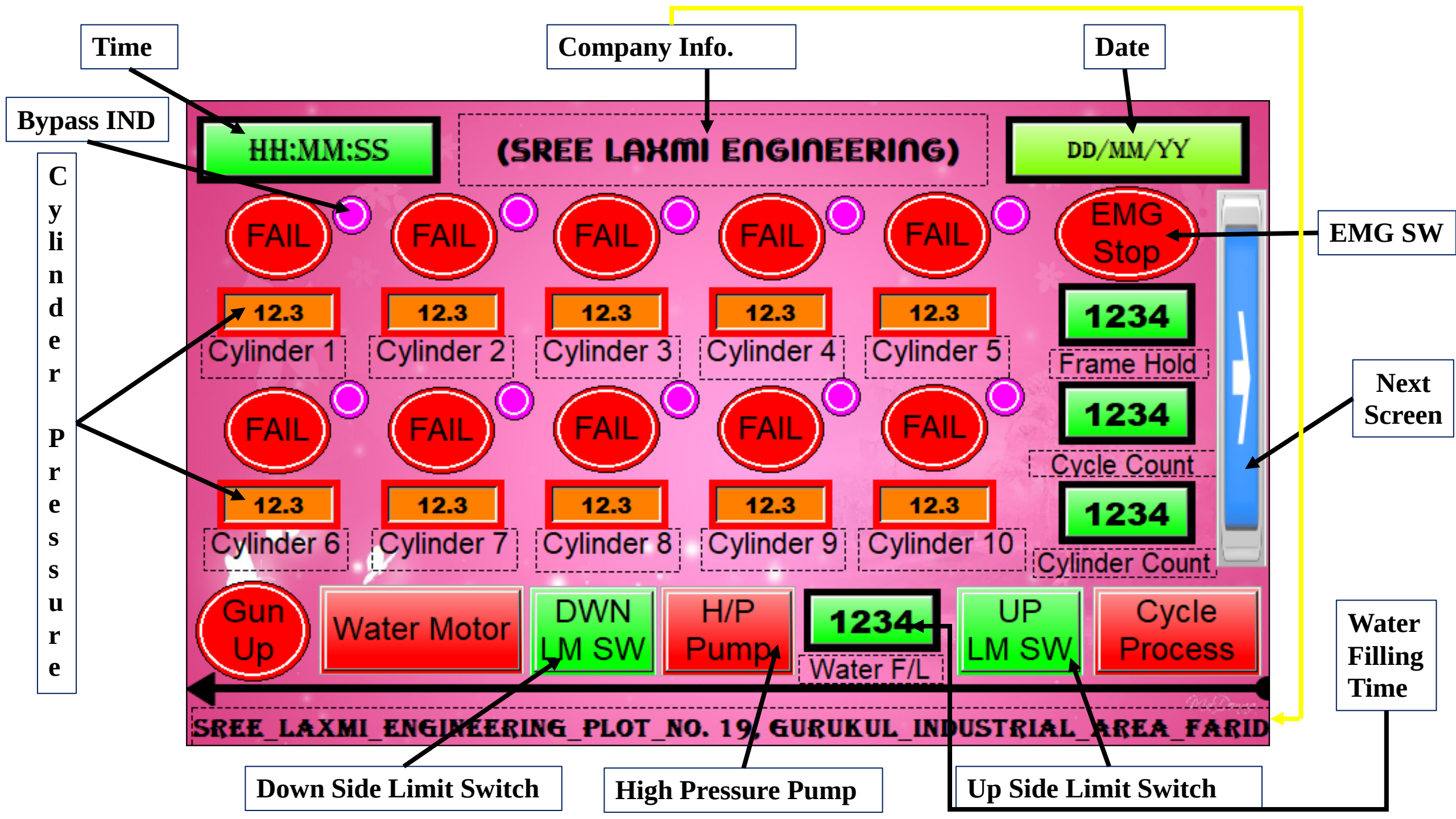
Y33	Spare
Y34	Green Light
Y35	Yellow Light
Y36	Spare
Y37	Water Motor



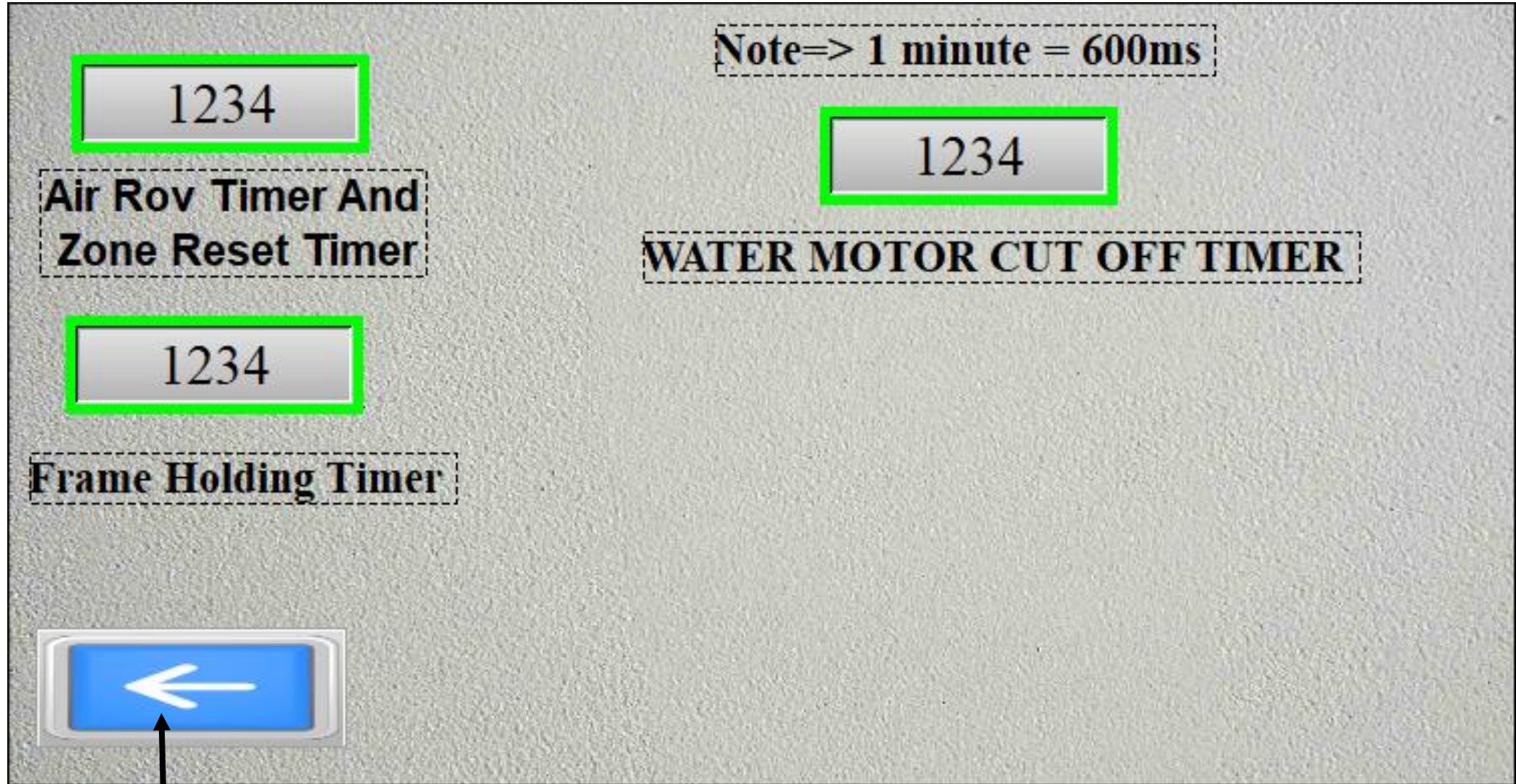


HMI

(Human Machine Interface)



Timer Adjustment



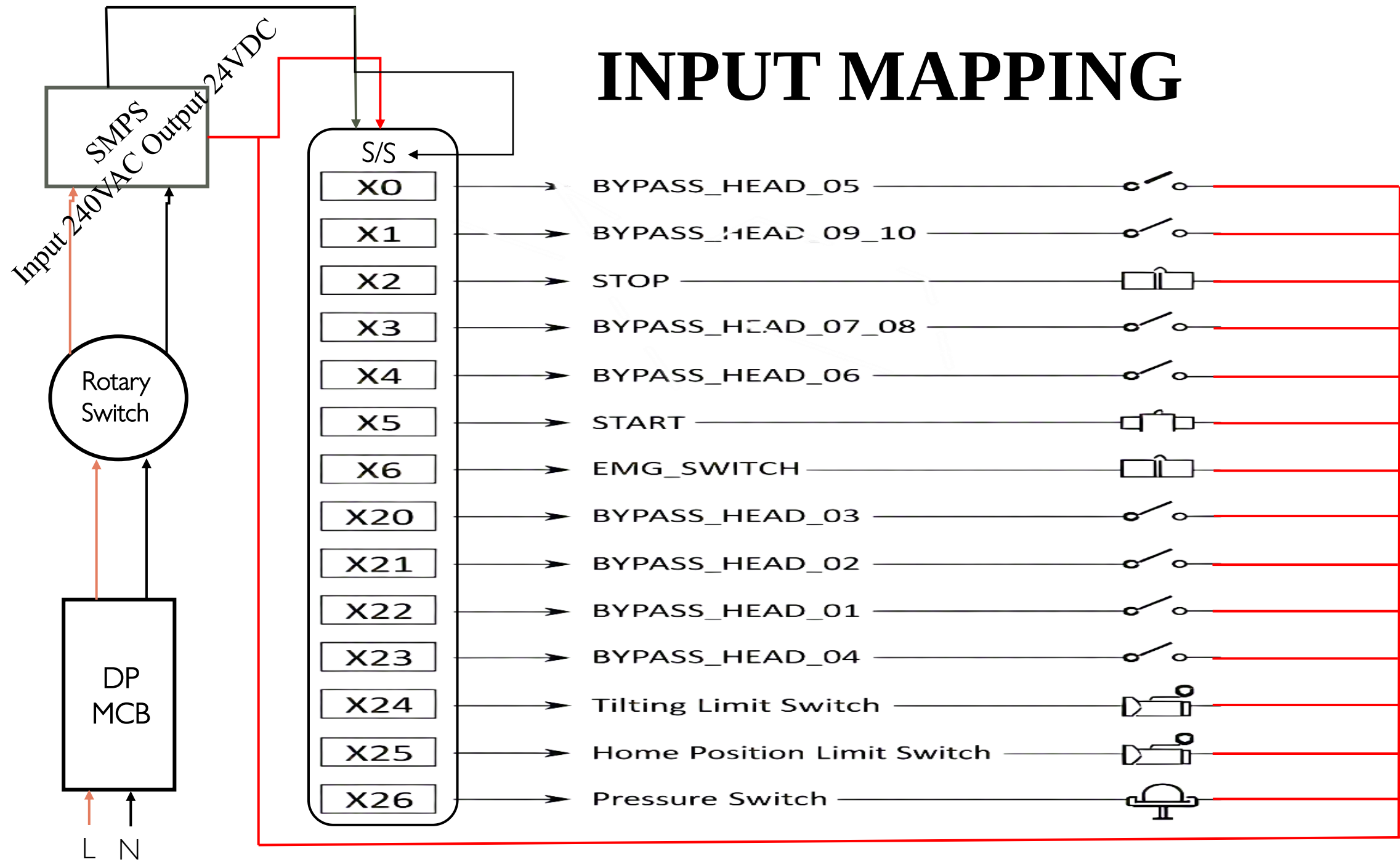
Back Screen



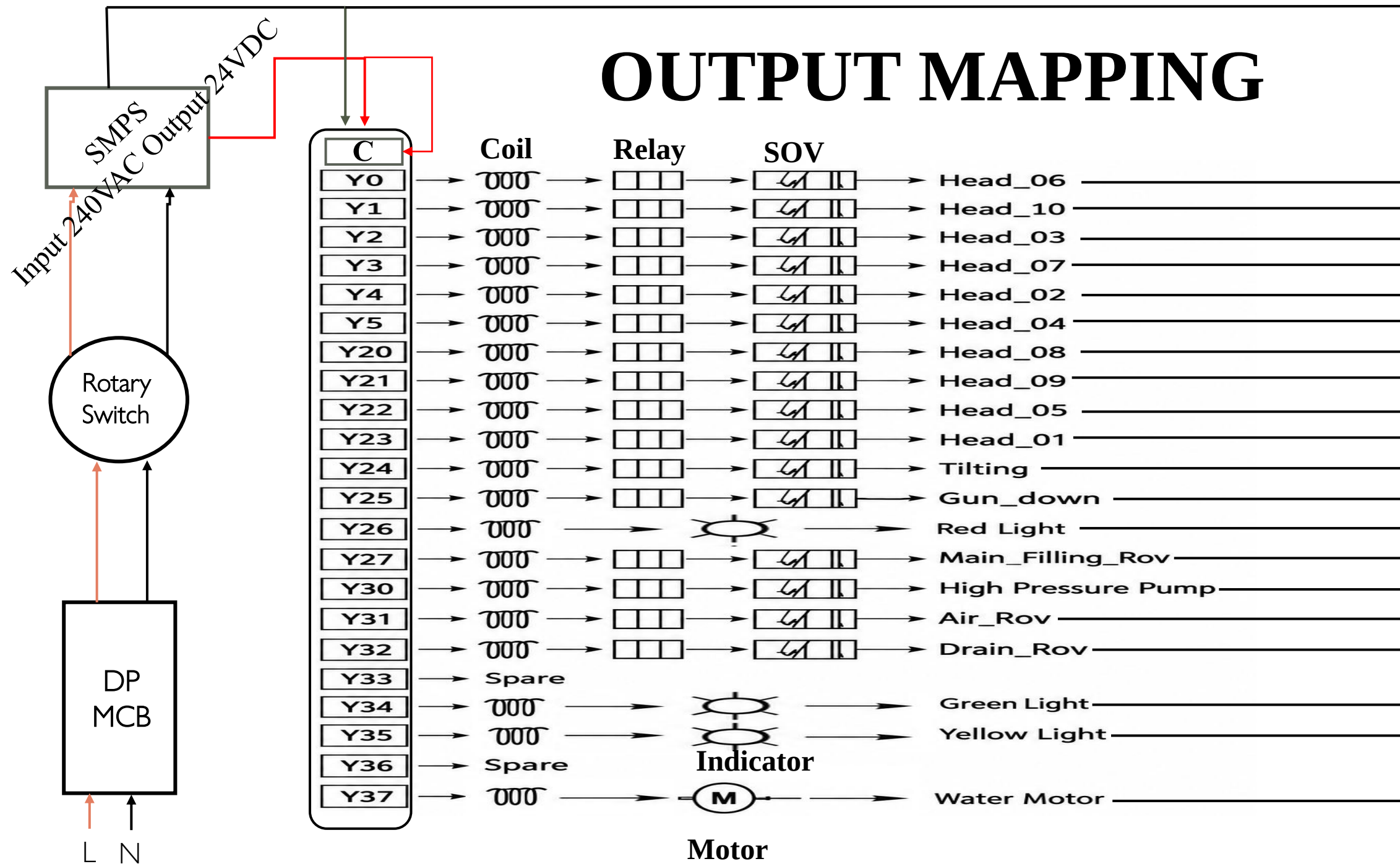
SLD (Single Line Diagram)

SLD Done By :- Somnath Vishwakarma

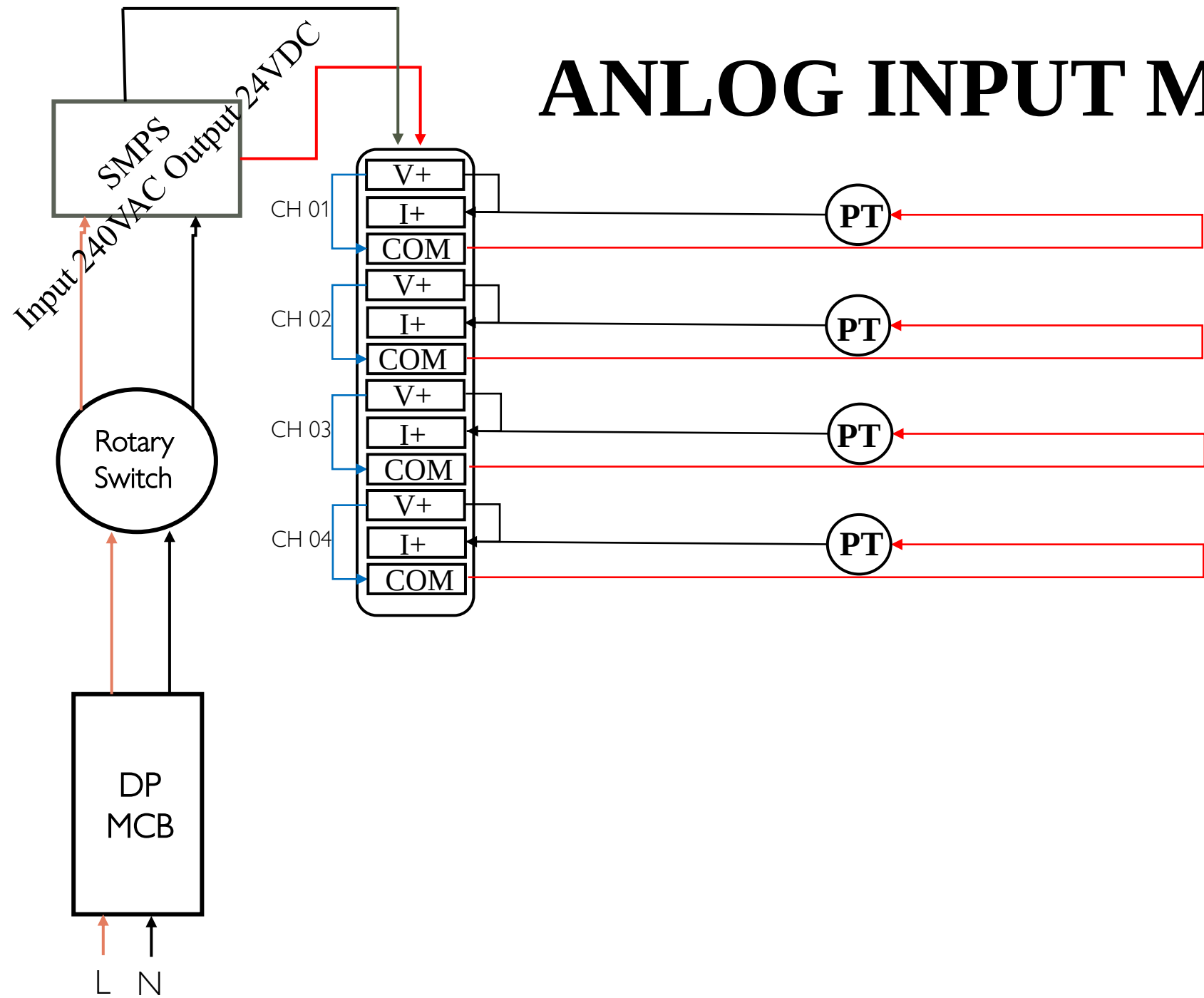
INPUT MAPPING



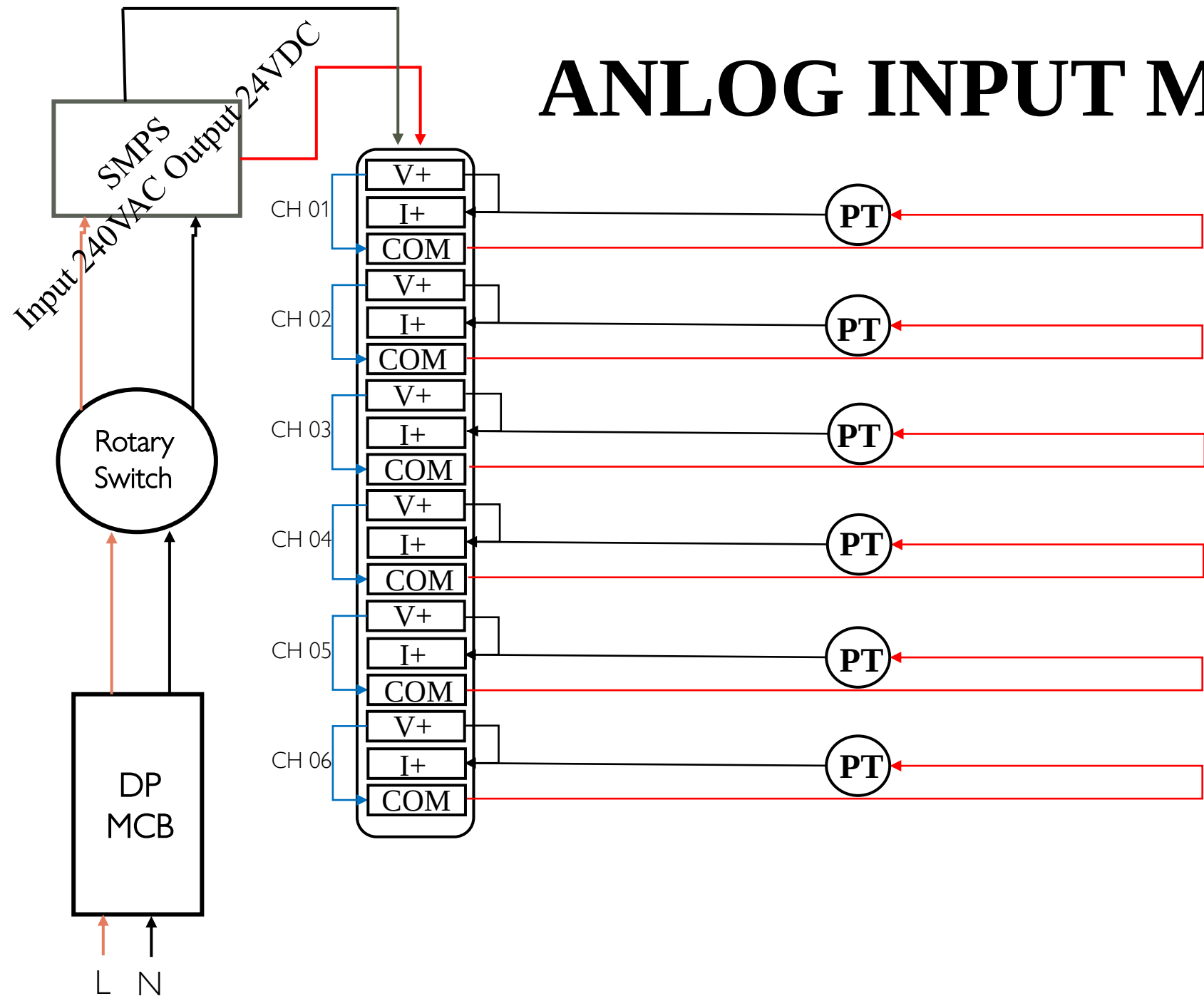
OUTPUT MAPPING



ANLOG INPUT MAPPING



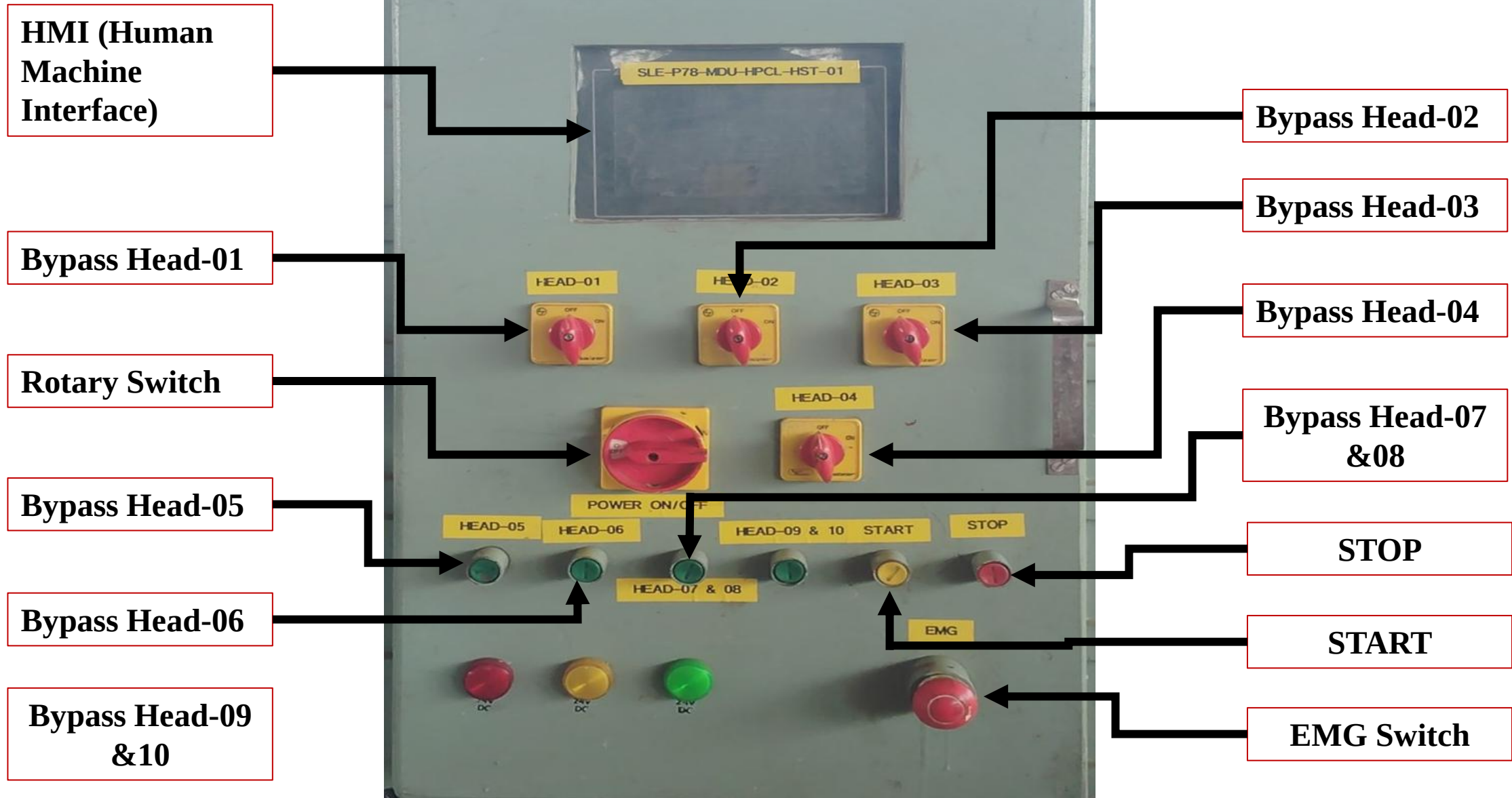
ANLOG INPUT MAPPING

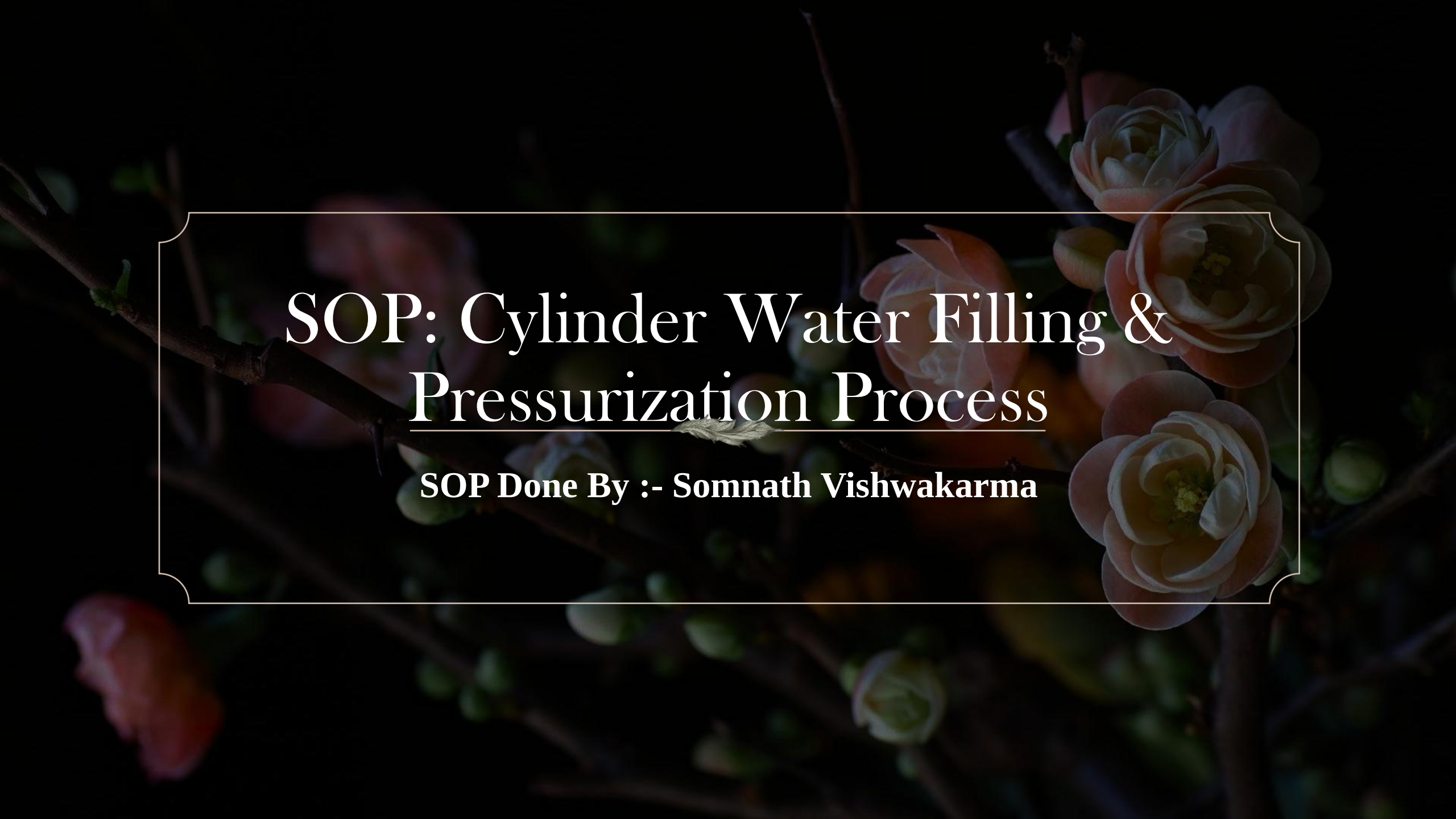




Panel Details (Make :- Saiex)

Panel Done By :- Somnath Vishwakarma





SOP: Cylinder Water Filling & Pressurization Process

SOP Done By :- Somnath Vishwakarma

Step	Action	Condition	Timing
1	Press Start Button → All head guns down	All cylinders loaded	Immediate
2	Align guns manually using hand lever	If any gun not centered	Until aligned
3	Press Start Button again	All guns aligned	Immediate
4	Turn ON Main Water Filling ROV	Start sequence	Immediate
5	Turn OFF Water Drain ROV	During filling	Immediate
6	Turn ON Filling Motor	Filling process	1500 Ms
7	Turn OFF Filling Motor & Main Water Filling ROV	After 1500 Ms	Immediate
8	Turn ON High Pressure Pump	After filling	Immediate
9	Build pressure to 25 Kg/cm ²	Normal operation	Until set point
10	Press Bypass Switch	If leakage at any head gun	Immediate
11	Turn OFF head ROVs one by one	When all heads reach 25 Kg/cm ²	Sequential
12	Retry pressure build	If any head not pressurized	Until set point
13	Frame hold	On pressure switch signal	300 Ms
14	Frame tilt reverse	After hold	180°
15	Activate Drain Process	On Down Side Limit Switch	Immediate
16	Open all head ROVs & Water Drain ROV	Drain sequence	Continuous
17	Pulse Air ROV ON/OFF	During drain	ON 50 mms / OFF 100 MS (repeat)
18	Continue drain cycle	Until Zone Reset Time completes	As per set time
19	Frame tilt upward	After Zone Reset	Immediate
20	Turn ON head ROVs	On Home Position Limit Switch	Immediate
21	Guns delatch	After head ROV ON	100 Ms



Machine Operating Sequence

1.Initial Condition – Cylinder Loading.

Once all cylinders are loaded, press the **START button**. All head guns will move downward.

2.Head Gun Alignment.

If any head gun is not properly aligned with the center position, manually align it using the **manual hand lever** in the UP/DOWN direction.

Ensure that **all head guns are properly aligned and centered**.

3.Start the Process.

After confirming that all head guns are correctly aligned, press the **START button** again.

4.Water Filling Process

Immediately after pressing **START**:

- Turn **ON the Main Water Filling ROV**.
- At the same time, turn **OFF the Water Drain ROV**.
- Turn **ON the Filling Motor** for **1500 Ms**.
- After 1500 Ms., turn the Filling Motor **OFF immediately**.
- At the same time, turn the **Main Water Filling ROV OFF**.

5. High-Pressure Pump Operation.

Immediately after completion of the water filling process, turn **ON the High-Pressure Pump**.

The pump will build pressure up to the set pressure of **25 kg/cm²**.

6. Leakage / Bypass Condition.

If any leakage is detected from the head gun side during pressurization, immediately activate **Bypass Mode** by pressing the corresponding **BYPASS switch**.

All head bypass switches are **located and locked on the front side of the PLC panel**.

7. Pressurization of All Heads.

When all head guns reach the required pressure of **25 kg/cm²**, immediately turn **OFF the Head ROVs one by one**.

If any head gun does not reach the required pressure, the **High-Pressure Pump will continue/restart the pressure-building operation** according to the configured set point.

As soon as the required pressure signal is received from the **Pressure Switch**, the sequence will proceed to the next step.

8. Frame Holding and Tilting.

Once the required pressure is achieved, hold the frame in position for 300 Ms.

After 300 mms, the frame will tilt 180° in the reverse direction.

9. Water Drain Process.

When the frame reaches the downward position and the **Down-Side Limit Switch** is activated, immediately start the **Water Drain Process**.

During the drain process:

- Open all **Head ROVs**.
- Open the **Water Drain ROV**.
- Turn the **Air ROV ON** for 50 Ms.
- Turn the Air ROV **OFF** for 100 Ms.
- Then turn it **ON** again for 50 Ms.
- Turn it **OFF** again for 100 Ms.

This **50 Ms. ON / 100 Ms. OFF** air-pulsing sequence will continue according to the configured **Zone Reset Time**.

10. Frame Return to Home Position.

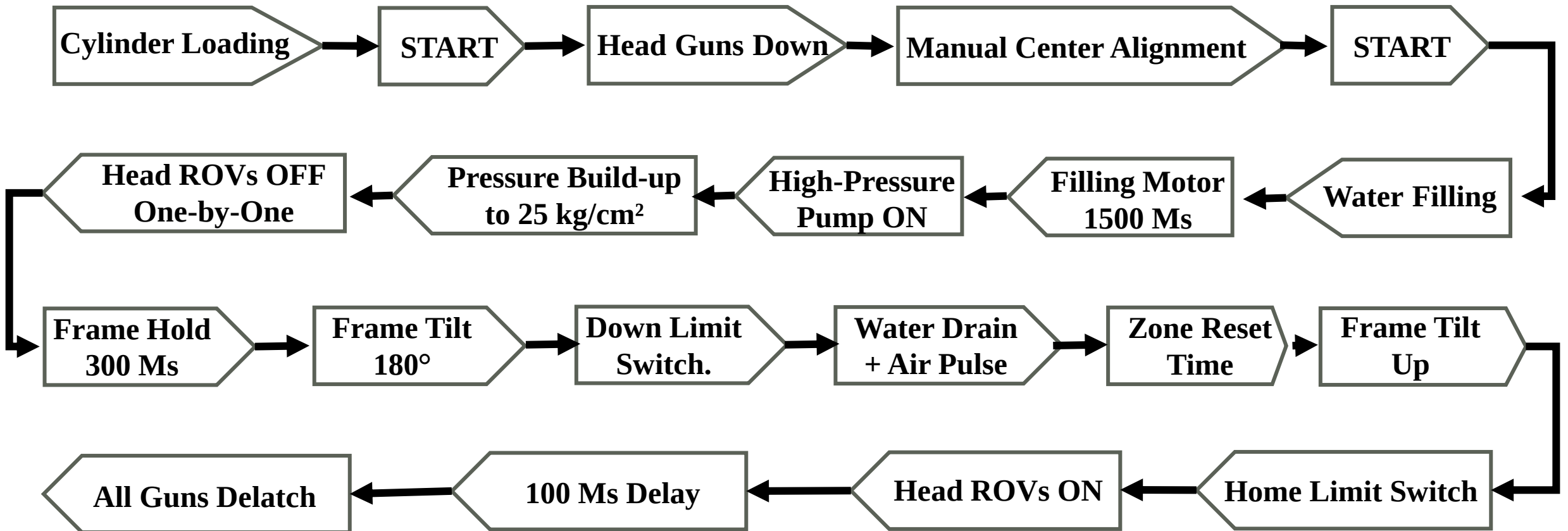
When the Zone Reset Time is completed, the frame will tilt **upward** and return to its home position.

11. Home Position and Gun Delatching.

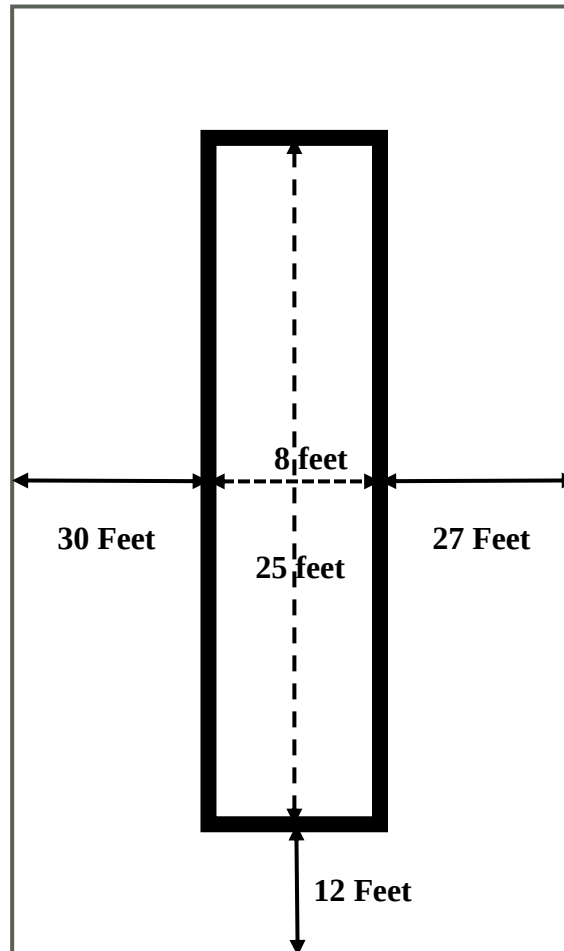
When the frame reaches the home position and the **Home Position Limit Switch** is activated:

- Turn **ON** the **Head ROVs**.
- After **100 Ms.**, all head guns will **delatch/release**.

Sequence Summary



HST Layout





Thank you

Warm Regards:- Somnath Vishwakarma

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Website:- [Project](#)

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